

Industrial Internship Report

Project Title: Banking Information System

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Internship: upskill Campus / UCT

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2. Preface

This internship provided practical exposure to software development using Core Java. I designed and implemented a Banking Information System using Object-Oriented Programming concepts. The project strengthened my problem-solving, programming, testing, and documentation skills.

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3. Introduction

The Banking Information System is a console-based Java application that automates common banking operations. The application uses classes, objects, encapsulation, ArrayList, and exception handling to manage customer accounts and transactions.

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4. About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt. Ltd. (UCT) works in digital transformation, IoT, cloud computing, machine learning, industrial automation, and software development. The internship introduced students to industry-oriented software development.

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5. About upskill Campus

upskill Campus provides career-focused internships, industrial projects, and skill development programs. It bridges academic learning with industry requirements.

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6. Objectives

- Learn Core Java
 - Apply OOP concepts
 - Develop a complete project
 - Practice GitHub version control
 - Improve software documentation skills
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7. Problem Statement

Manual banking operations are time-consuming and susceptible to errors. The objective is to develop a Java application that automates account creation, deposits, withdrawals, balance enquiry, and transaction history while providing a simple user interface.

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Code GitHub Link:

<https://github.com/BhargavHarshaN/upskillcampus/blob/main/BankingInformationSystem.java>

Report GitHub Link:

[https://github.com/BhargavHarshaN/upskillcampus/blob/main/BankingInformationSystem_Internship_Report%20\(2\).pdf](https://github.com/BhargavHarshaN/upskillcampus/blob/main/BankingInformationSystem_Internship_Report%20(2).pdf)

8. System Design

Modules:

- Account Creation
- Deposit
- Withdraw
- Balance Enquiry
- Account Details
- Transaction History

Data Flow:

User → Menu → BankingInformationSystem → BankAccount → Output

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9. High Level Diagram

[Insert High-Level Diagram Here]

User → Menu → BankingInformationSystem → BankAccount → Transaction
Processing → Output

[Insert High-Level Diagram Here]

User → Menu → BankingInformationSystem → BankAccount → Transaction
Processing → Output

10. Low Level Diagram

[Insert Low-Level Diagram Here]

Main() → Menu → Switch Case → Search Account → Perform Operation → Display Result

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11. Interfaces

Console Interface:

1. Create Account
2. Deposit
3. Withdraw
4. Check Balance
5. Account Details
6. Transaction History
7. Exit

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12. Performance Test

The application was tested with multiple accounts and transactions. The system produced correct balances and transaction histories while maintaining fast execution and low memory usage.

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13. Test Cases

1. Create Account – Pass
2. Deposit – Pass
3. Withdraw – Pass
4. Invalid Account – Pass
5. Balance Enquiry – Pass
6. Transaction History – Pass

1. Create Account – Pass
2. Deposit – Pass
3. Withdraw – Pass
4. Invalid Account – Pass
5. Balance Enquiry – Pass
6. Transaction History – Pass

14. My Learnings

This project enhanced my understanding of Java programming, OOP, collections, exception handling, software testing, debugging, GitHub, and technical documentation.

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15. Future Scope

Future enhancements include MySQL integration, Swing/JavaFX GUI, login authentication, password encryption, fund transfer, account deletion, and online banking services.

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